1. DEFINITION

DEFINITION
Caprylocaproyl macrogol-8 glycerides EP / Caprylocaproyl polyoxyl-8 glycerides NF

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Caprylocaproyl Polyoxylglycerides or: PEG-8 caprylic/capric glycerides
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	00BT03FSO2
ADDITIONAL INFORMATION ON THE DENOMINATION	Polyethylene glycol 400 caprylate/caprate glycerides Polyoxylethylene (8) caprylate/caprate glycerides Polyoxylethylene caprylate/caprate glycerides Caprylocaproyl Polyoxylglycerides 400 PEG-8 Caprylic/Capric Glycerides
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Alcoholysis reaction between PEG-8 and Medium-chain Triglycerides (C8 - C10) from vegetable oil

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COMPOSITION	
Well defined multi-constituent substance composed of mono-, di- and triglycerides and PEG-8 mono- and di- esters of Caprylic/Capric acids (C8 - C10)	
Full Labeling (Key letters system) **$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$**	
Caprylocaproyl Polyoxylglycerides (Or) PEG-8 caprylic/capric glycerides	$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)	
None	

4. RSPO CERTIFICATION

ROUNDTABLE ON SUSTAINABLE PALM OIL http://www.rspo.org/certification	
Product status	**Gattefossé RSPO certification number**
Product certified RSPO Mass Balance. More information on: www.RSPO.org	Supply Chain Certificate (SCC) number: BVC-RSPO-1-1106949778

5. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	5734
Canada MF (Type III – Excipient)	2006-034 (inactive)
For submission of the DMF in another country, please contact your local Gattefossé representative	

6. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	**Conforms to monograph**
European Pharmacopoeia (Ph.Eur.) Current edition	Caprylocaproyl macrogol-8 glycerides
USP/ NF Current edition	Caprylocaproyl polyoxyl-8 glycerides

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7. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓		✓	✓ (6)	✓	✓ (8) 21CFR180.910 21CFR180.930 Uses: emulsifiers

EUROPEAN REGULATION
(1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption)
(2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units)
(3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether)
(4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin
USA REGULATION
- Veterinary drugs
FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety
(5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR)
(6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database)
- Pesticides
(8)EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)

8. SPECIFIC PHARMACEUTICAL REGULATIONS

CHINA BUNDLING REVIEW			
Generic name	**Registration N°**	**Company**	**CFDA/CDE Reference (http://www.cde.org.cn/yfb.do?method=main)**
Caprylocaproyl macrogolglycerides 辛酸癸酸聚乙二醇甘油酯	F20190001267	Gattefossé SAS	Raw material approved for use in marketed formulations in Joint-Review with drug applications.
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

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REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - [Australian Register of Therapeutic Goods Ingredients]			
Status	**Ingredient Name**	**Use**	**Restrictions**
Approved	Caprylocaproyl macrogolglycerides	Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Prescription Medicines Not available as an Equivalent Ingredient in any application	/
Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homoeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA [NHPID]		
Status	**NHPID name**	**Route of administration/ Restrictions**
Approved non-medicinal ingredient	PEG-8 Caprylic/Capric Glycerides	Role Non-medicinal : Purposes: Emulsifying Agent Route Of Administration: For topical use only. When formulated to be non-irritating.
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

9. SAFETY ASSESSMENT

SAFETY STUDIES PERFORMED BY GATTEFOSSÉ					
Study type/ duration	**Route**	**Species**	**Test article**	**Results/Conclusions**	**Reference**
ADME	IV (bolus) Oral (gavage)	Rat	C^{14}-LABRASOL® (vehicle: 0.9% NaCl) IV: 10 mg/kg (single dose) Oral: 10 and 150 mg/kg (single dose)	Please consult study summary.	(Gattefossé data, GAT/PEG/99001)
Acute irritation (JORF)	Dermal	Rabbit	LABRASOL® Pure (undiluted)	Non Irritant Cutaneous irritation index:0.00	(Gattefossé data, GAT-72105, 1972)
Acute irritation (Patch test)	Dermal	Human	LABRASOL® Pure (undiluted)	Well Tolerated Cutaneous irritation index: 0.00	(Gattefossé data, GAT-92108, 1992)
Acute irritation (JORF)	Ocular	Rabbit	LABRASOL® Pure (undiluted)	Non Irritant Ocular irritation index (max 72h): 10.0	(Gattefossé data, GAT-72105, 1972)
Acute toxicity (OECD 401)	Oral	Rat	LABRASOL® Pure (undiluted) Dose levels : 20, 22.4, 25.1, 28.21 and 31.60 g/kg	$LD50_{(oral)}$ > 22000 mg/kg	(Gattefossé data, GAT-89107, 1989)
Acute toxicity (Dose escalating)	IV bolus (tail vein)	Mouse	LABRASOL® Vehicle: Physiological saline solution M: 25, 50, 100, 200, 400, 300, 250, 150 and 125 mg/kg F: 25, 50, 100, 200, 400, 300 and 250 mg/kg	$MTD_{(IV)}$ (M): 125 mg/kg $MTD_{(IV)}$ (F): 250 mg/kg	(Gattefossé data, GAT-MTDs-PH-08/0475, 2009)

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SAFETY STUDIES PERFORMED BY GATTEFOSSÉ					
Study type/ duration	**Route**	**Species**	**Test article**	**Results/Conclusions**	**Reference**
Acute toxicity (eq OECD 423)	IV bolus (tail vein)	Mouse	LABRASOL® Vehicle: Physiological saline solution M: 125 mg/kg F: 250 and 125 mg/kg	$MTD_{(IV)}$: 125 mg/kg F: Administration at 250 mg/kg induces mortality (2/2) M+F: Administration at 125 mg/kg induces no mortality and toxicity signs.	(Gattefossé data, GAT-TAIs-PH-08/044, 2009)
Subchronic 14-day toxicity study (DRf)	Oral (Gavage)	Rat	LABRASOL® Vehicle/control: 0.9% NaCl for injection Dose levels: 100, 300, 1000 and 3000 mg/kg/day	$NOAEL_{(oral)}$ > 3000 mg/kg/day	(Gattefossé data, GAT- WIL261006, 1999)
Subchronic 6-month toxicity study	Oral (Gavage)	Rat	LABRASOL® Vehicle/control: 0.9% NaCl for injection Dose levels: 300, 1000 and 3000 mg/kg/day	$NOEL_{(oral)}$: 300 mg/kg/day $NOAEL_{(oral)}$ > 3000 mg/kg/day	(Gattefossé data, GAT- WIL261005, 2001)
Subchronic 14-day toxicity study (DRF)	Oral (Capsule)	Dog	LABRASOL® Vehicle: None Dose levels: 100, 300, 1000 and 3000 mg/kg/day	In high-dose group, moderate suppurative inflammation of the lungs. No adverse Effects on survival and clinical observations.	(Gattefossé data, GAT- WIL261003, 1996)
Subchronic 13-week toxicity study	Oral (Capsule)	Dog	LABRASOL® Vehicle: None Dose levels: 300, 1000 and 3000 mg/kg/day	$NOEL_{(oral)}$: 1000 mg/kg/day $NOAEL_{(oral)}$ > 3000 mg/kg/day	(Gattefossé data, GAT- WIL261004, 1997)
Suchronic 3-month toxicity study (JORF)	Dermal	Rabbit	LABRASOL® Pure (undiluted)	$NOAEL_{(dermal)}$ > 2000 mg/kg/day Very Well Tolerated	(Gattefossé data, GAT-72105, 1972)
Reprotoxicity 7-day study (DRF-Segment II)	Oral (Gavage)	Rat	LABRASOL® Vehicle/control: purified water Dose levels: 0, 1000, 2000 and 3000 mg/kg/day	$NOAEL_{Matern}$ > 3000 mg/kg/day $NOAEL_{Embryo/fetal}$ > 3000 mg/kg/day No signs of embryotoxicity, fetotoxicity or teratogenicity were noted at any dose-level.	(Gattefossé data, GAT-CIT-22667, 2002)
Reprotoxicity study Embryo-fetal development (Segment II)	Oral (Gavage)	Rat	LABRASOL® Vehicle/control: purified water Dose levels: 0, 1000, 2000 and 3000 mg/kg/day	$NOAEL_{Matern}$: 2000 mg/kg/day $NOAEL_{Embryo/fetal}$: 3000 mg/kg/day No signs of embryotoxicity, fetotoxicity or teratogenicity were noted at any dose-level.	(Gattefossé data, GAT-CIT-22668, 2002)
Ames test (OECD 471)	In vitro	S. typhimurium	LABRASOL®	Negative. Not mutagenic.	(Gattefossé data, GAT-Pharmak 62695, 1996)
Mouse Lymphoma Assay	In vitro	Mouse lymphoma cell	LABRASOL®	Negative.	Gattefossé data, GAT-Chry 0313FG18.002, 1998)
Mammalian Erythrocyte Micronucleus Test (OECD 474)	IP	Mouse	LABRASOL®	Negative. Not clastogenic.	(Gattefossé data, GAT-Chry 935/012, 1998)
DNA damage and repair (UDS: Unscheduled DNA synthesis in mammali)	Oral	Rat	LABRASOL®	Not Genotoxic for dose levels up to 5000 mg/kg/day	Gattefossé data, GAT-Pasteur IPL-R 010501, 2001)

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CIR COMPENDIUM (Cosmetic Ingredient Review) (http://www.cir-safety.org)			
Denomination	**Reference**	**Maximum "as used" concentration**	**Conclusions**
PEG-8 Caprylic/Capric Glycerides (PEGylated Alkyl Glycerides)	2015, Jan 13	PEG-6 Caprylic/Capric Glycerides : up to 5 % PEG-7 Caprylic/Capric Glycerides : up to 2 % PEG-8 Caprylic/Capric Glycerides : None reported	On the basis of the animal and clinical data included in this report, the CIR Expert Panel concludes that PEGylated Alkyl Glycerides are safe as used in cosmetics.

10. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
	CAS #	**EC #**	**Chemical name**
Caprylocaproyl Polyoxylglycerides	61791-29-5 or [73398-61-5 (or 85409-09-2) + 223129-75-7]	Not applicable or [277-452-2 (or 287-075-5) + Not applicable]	Fatty acids, coco, ethoxylated or [Glycerides, mixed decanoyl and octanoyl (or Glycerides, C8-10) + Poly(oxy-1,2-ethanediyl), α-hydro-ω-hydroxy-, mixed decanoate and octanoate]
EC numbers: There are three separate inventories. These are the European Inventory of Existing Commercial Chemical Substances (EINECS), the European List of Notified Chemical Substances (ELINCS) and the No-Longer Polymers (NLP) list. EINECS numbers start with 2 or 3. ELINCS numbers start with 4. NLP numbers start with 5. Substances without EC numbers were assigned "list numbers" in order to facilitate REACh submissions. They are solely of administrative, and not of regulatory relevance. The list numbers start with 6, 7, 8 or 9.			

NATIONAL INVENTORIES					
	U.S.A. (TSCA)	**Canada (DSL/NDSL)**	**Australia (AIIC)**	**China (IECSC)**	**Other inventories**
Caprylocaproyl Polyoxylglycerides	Listed	DSL listed	Listed	Listed	Fatty acids, coco, ethoxylated: AIIC, AREC, DSL, ECL, IECSC, NZIoC, PICCS, REACH, TCSI, TSCA, VNECI (or) [Glycerides, mixed decanoyl and octanoyl: AIIC, AREC, DSL, ECL, EINECS, IECSC, NZIoC, PICCS, REACH, TCSI, TSCA, TDCA/ Ester of decanoic acid, octanoic acid and glycerol: VNECI (or) Glycerides, C8-10: AIIC, DSL, EINECS, NZIoC, PICCS, REACH, TCSI, AREC, ECL, IECSC) + Poly(oxy-1,2-ethanediyl), α-hydro-ω-hydroxy-, mixed decanoate and octanoate: NDSL, TCSI, TSCA, VNECI]
U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics… are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.					

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NATIONAL INVENTORIES					
	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Other inventories
Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation. Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.					

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
	Status of the product	Status per substance
Caprylocaproyl Polyoxylglycerides	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACh regulation.

11. <u>HISTORY OF USE</u>

11.1. Pharmaceutical applications

11.1.1. FDA IIG reference for Chemical equivalent

FDA INACTIVE INGREDIENT GUIDE (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
CAPRYLOCAPROYL POLYOXYLGLYCERIDES 8 (UNII: 00BT03FSO2)	ORAL	CAPSULE	/	1648 mg
	ORAL	CAPSULE, LIQUID FILLED	/	3623mg
	ORAL	SOLUTION	61.2mg/1ml	/
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

11.1.2. FDA IIG references for larger family of chemicals

The following reference covers multiconstituant substances containing PEG-8 caprylic/capric glycerides

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FDA Inactive ingredient guide (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm) Update : 27/02/2020		
Inactive ingredient	**Route / Dosage form**	**Maximum Potency per unit dose**
PEG VEGETABLE OIL (UNII: Not available)*	*IM – SC (Injection)*	*7 %*
	IV(INFUSION) (Injection)	*5 %*
Maximum potency: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the "maximum potency" field will be blank.		

**removed from IIG in July 2019 due to new data standard. The reference was not removed from IIG due to safety concerns.*

11.1.3. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENTS (Current edition by Paul J. Sheskey, Bruno C Hancock, Gary P Moss Dabid J Goldfard)			
Denomination	**Functional Category**	**Regulatory status**	**Safety**
Polyoxylglycerides	Dissolution enhancer; emulsifying agent; nonionic surfactant; penetration agent; solubilizing agent; sustained release agent.	Lauroyl polyoxylglycerides and stearoyl polyoxylglycerides are approved as food additives in the USA. Included in the FDA Inactive Ingredients Database (oral route: capsules, tablets, solutions; topical route: emulsions, creams, lotions; vaginal route: emulsions, creams). Oleyl polyoxylglycerides are included in a topical cream formulation licensed in the UK.	Polyoxylglycerides are used in oral and topical pharmaceutical formulations, and also in cosmetics and food.

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